

Q₁:

$$(1) \quad x(n) = 2 \cos(\pi n)$$

$$\text{Discrete period } N = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2$$

synthesis equation in discrete time:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi}{N} kn}$$

$$= \sum_{k=0}^1 a_k e^{j \pi kn}$$

$$= a_0 e^0 + a_1 e^{j \pi n}$$

$$= a_0 + a_1 e^{j \pi n}$$

by Euler's formula:

$$e^{j\theta} = \cos\theta + j \sin\theta$$

when $\theta = \pi n$:

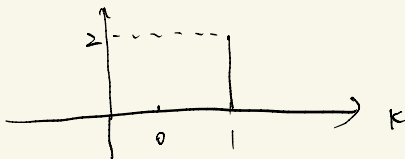
$$e^{j \pi n} = \cos(\pi n) + j \sin(\pi n)$$

$$x[n] = a_0 + a_1 \cos(\pi n) + a_1 j \sin(\pi n)$$

$$\because x(n) = 2 \cos(\pi n) \quad \sin \pi n = 0$$

$$\therefore a_1 = 2 \quad a_0 = 0$$

$$\therefore \text{coefficients: } a_0 = 0 \quad a_1 = 2$$



$$(z), \quad x(n) = -3 \sin\left(\frac{\pi}{12}n\right)$$

$$\text{Discrete period } N = \frac{2\pi}{\frac{\pi}{12}} = 24$$

synthesis equation:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{N}kn}$$

$$N=24$$

$$x[n] = \sum_{k=0}^{23} a_k e^{j\frac{k\pi}{12}n}$$

$$= a_0 e^0 + a_1 e^{j\frac{\pi}{12}n} + a_2 e^{j\frac{2\pi}{12}n} + \dots + a_{23} e^{j\frac{23\pi}{12}n}$$

by Euler's formula:

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\sin\frac{\pi}{12}n = \frac{e^{j\frac{\pi}{12}n} - e^{-j\frac{\pi}{12}n}}{2j}$$

$$x(n) = \frac{-3}{2j} (e^{j\frac{\pi}{12}n} - e^{-j\frac{\pi}{12}n})$$

$$= -\frac{3}{2j} e^{j\frac{\pi}{12}n} + \frac{3}{2j} e^{-j\frac{\pi}{12}n}$$

$$\text{when } k=1 \quad e^{j\frac{k\pi}{12}n} = e^{j\frac{\pi}{12}n}$$

$$\text{when } k=-1 \quad e^{j\frac{k\pi}{12}n} = e^{-j\frac{\pi}{12}n}$$

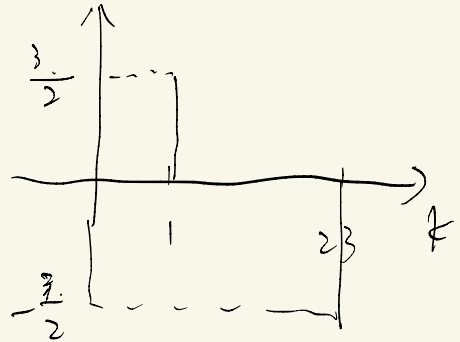
$$\because -1 \equiv 23 \pmod{24}$$

$$\therefore k=23 \quad e^{j\frac{k\pi}{12}n} = e^{-j\frac{\pi}{12}n}$$

$$\therefore a_1 = -\frac{3}{2j} = \frac{3j}{2} \quad a_{23} = \frac{3}{2j} = -\frac{3j}{2} \quad \checkmark$$

$a_k = 0$ for other k 's

magnitude spectrum:



$$(3) x(n) = \cos\left(\frac{2\pi n}{5}\right) - 2 \sin\left(\frac{\pi n}{12}\right)$$

$$N_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{\frac{2\pi}{5}} = 5$$

$$N_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{\frac{\pi}{12}} = 24$$

$$N = \text{lcm}(5, 24) = 120$$

$$x(n) = \frac{1}{2} e^{\frac{2\pi j n}{5}} + \frac{1}{2} e^{-\frac{2\pi j n}{5}} + j e^{\frac{\pi j n}{12}} - j e^{-\frac{j \pi n}{12}}$$

$$\text{let } \frac{\pi}{60} j n k = \pm \frac{2\pi n j}{5} \quad k = \pm 24$$

$$\text{let } \frac{\pi}{60} j n k = \pm \frac{j \pi n}{12} \quad k = \pm 5$$

synthesis formula:

$$x(n) = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi}{N} k n} = a_0 + a_1 e^{j \frac{\pi}{60} j n} + a_2 e^{2 \times \frac{\pi}{60} j n} + \dots + a_{119} e^{119 \times \frac{\pi}{60} j n}$$

by Euler's formula:

$$\cos \theta = \frac{e^{j\theta} - e^{-j\theta}}{2}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos \frac{2\pi n}{5} = \frac{e^{j \frac{2\pi n}{5}} + e^{-j \frac{2\pi n}{5}}}{2}$$

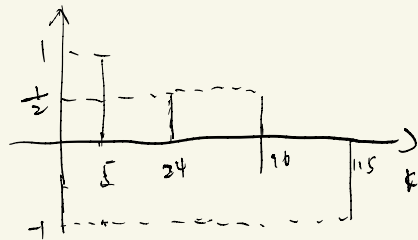
$$2 \sin \frac{\pi n}{12} = \frac{e^{j \frac{\pi n}{12}} - e^{-j \frac{\pi n}{12}}}{j}$$

$$\therefore a_{24} = \frac{1}{2} \quad -24 \equiv 96 \pmod{120}$$

$$a_{96} = \frac{1}{2}$$

$$a_5 = j \quad -5 \equiv 115 \pmod{120}$$

$$a_{115} = -j$$



$$= -j e^{\frac{\pi j n}{12}} + j e^{-\frac{j \pi n}{12}}$$